Smart Internz

Project Initialization and Planning Phase

Date	07 October 2025
Team ID	Xxxxxx
Project Name	Predicting Plant Growth Stages with Environmental and Management Data Using Power BI
Maximum Marks	3 Marks

Define Problem Statements (Customer Problem Statement Template):

In modern agriculture, understanding the factors that influence plant growth is crucial for achieving higher yield and sustainability. Farmers, researchers, and agricultural organizations often collect large amounts of data related to sunlight, temperature, humidity, and soil nutrients. However, this data remains underutilized due to the difficulty of identifying patterns and relationships among these variables.

The challenge lies in accurately **classifying plant growth stages** and understanding how different environmental conditions affect overall plant health. Manual observation is time-consuming, error-prone, and inconsistent. Without proper insights, farmers may over- or under-water their crops, apply incorrect fertilizers, or miss early signs of poor growth conditions.

Section	Description
I am	a small-scale farmer who depends on environmental conditions like sunlight, soil quality, temperature, and humidity to ensure healthy plant growth
I'm trying to	understand how different environmental factors affect the growth stages of my plants and identify what conditions lead to poor or healthy growth
But	I find it difficult to track and analyze all these variables manually because the data is

	large, scattered, and hard to interpret without technical expertise
Because	I lack the right tools or systems to convert raw environmental data into meaningful insights that can guide my farming decisions
Which makes me feel	frustrated, uncertain, and worried about making incorrect choices that might harm my crops or reduce yield

Reference: https://miro.com/templates/customer-problem-statement/

Example:

Problem Statement (PS)	**I am (Customer)**	**I'm trying to**	**But**	**Because**	**Which makes me feel**
PS-1	a farmer/researcher	understand which factors (like sunlight, temperature, or humidity) affect the growth of my plants the most	I can't easily track or analyze all these conditions manually	the data I collect daily from my farm is too large and confusing to interpret without technical help	frustrated and uncertain about how to make the right decisions for better plant growth
PS-2	an agricultural scientist	classify and predict plant growth stages	it's challenging to identify clear	traditional methods take too	limited and eager for a smarter,

		using real-world data	patterns among multiple growth variables like soil moisture, light intensity, and nutrient levels	much time and often fail to generalize across different plant species	data-driven solution that can help discover insights faster and more accurately